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The Impact of VTubers and Streamers on the Purchase Intention of Otaku and Non-Otaku Respondents: A Comparative Study

Smith Boonchutima and Apinya Surakanon

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ABSTRACT

This study aims to examine the impact of two types of influencers, VTubers and streamers, on the purchase intention of Otaku and non-Otaku respondents. A quantitative approach using an online survey was conducted among Thai respondents ($N = 686$) to assess the influence of expertise, attractiveness, and trustworthiness on their purchase intention induced by VTubers and streamers. The results indicate that expertise, attractiveness, and trustworthiness are significant predictors of purchase intention for both types of influencers. However, the impact of these predictors varies depending on the type of influencer and respondent. The findings contribute to the literature on influencer marketing, source credibility, and Otaku subculture and have implications for marketers and content creators in the influencer marketing industry.

The rapid growth of social media platforms and online gaming has given rise to a new breed of influencers in the gaming industry. These influencers, consisting of both live streamers and virtual YouTubers (VTubers) (Berryman et al., 2021), have gained significant popularity and have been able to impact the purchasing decisions of their followers. As gaming influencers gain more prominence, it becomes increasingly important to understand the factors that contribute to their credibility and how these factors influence audience behavior, particularly in terms of purchase intentions (Djafarova & Rushworth, 2017).

One subculture that has shown a high level of engagement with gaming influencers is the Otaku community. Otaku is a term often used to describe individuals who are passionate about various aspects of Japanese popular culture, such as anime, manga, and video games (Kinsella, 1998). Given the strong interest in gaming among the Otaku community, it is crucial to explore the relationship between gaming influencer credibility and purchase intentions within this context.

This study aims to investigate the relationship between female gaming influencer credibility and purchase intention among Otaku and non-Otaku groups. This paper will first review the relevant literature on influencer credibility, the Otaku subculture, and audience engagement. Next, the methodology and measures used in this study will be described, followed by the presentation of results and discussion. Last, the paper will address limitations, implications, and suggestions for future research.

Literature review

Influencer credibility

Credibility is a vital factor in persuasion and communication, as it influences the audience's perception and acceptance of the information being shared (Hovland et al., 1953). The concept of credibility is multifaceted, encompassing aspects such as expertise, trustworthiness, and attractiveness (Ohanian, 1990). When assessing the credibility of an influencer, audiences often evaluate these dimensions to determine the persuasiveness and reliability of the information being provided (Ohanian, 1990).

Source credibility theory (Hovland et al., 1953) posits that the credibility of a communicator or source directly impacts the persuasiveness of their message. This theory has been applied to various

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communication contexts, including advertising (Baker & Churchill, 1977), marketing (Kotler et al., 2010) and, more recently, social media influencers (Alkan & Ulas, 2023). According to this theory, when an audience perceives an influencer as credible, they are more likely to accept the information provided, which can influence their behavior and decision-making processes (Petty & Cacioppo, 1986).

Ohanian (1990) identified three key dimensions of credibility: expertise, trustworthiness, and attractiveness. Expertise refers to the perception that an influencer has the knowledge, experience, or skills necessary to provide reliable information (Afifah, 2022; Ohanian, 1990). Trustworthiness is the audience's belief in the influencer's honesty and sincerity, while attractiveness encompasses the influencer's physical appearance and overall likeability (Afifah, 2022; Ohanian, 1990).

Previous research has demonstrated the importance of these dimensions in influencer marketing. For instance, Lou and Yuan (2019) found that an influencer's expertise and trustworthiness positively impacted audience purchase intentions, while attractiveness played a significant role in enhancing audience engagement. Similarly, a study by Djafarova and Rushworth (2017) revealed that credibility dimensions affected the audience's perception of the influencer and their willingness to act upon the recommendations provided.

Other studies have also identified attractiveness as an important credibility dimension. Boonchutima and Sankosik (2022) found that an influencer's attractiveness could affect consumer attitudes and purchase intentions. Similarly, a study by Park et al. (2022) found that perceived attractiveness of an influencer could significantly affect purchase intentions and willingness to pay.

The Otaku subculture

The term "Otaku" originally referred to an obsessive fan of anime and manga, but it has since evolved to include anyone with a strong interest in Japanese popular culture, including video games. Otaku culture is characterized by a strong sense of community, with fans often gathering at events such as anime and gaming conventions. Otaku are known for their passion and dedication to their hobbies, often spending large amounts of time and money on their interests (Kinsella, 1998).

The Otaku community has a strong connection to the gaming industry, as many popular video games have roots in Japanese culture and media. Games such as Final Fantasy and Pokemon, which have gained worldwide popularity, originated in Japan and have a significant following among the Otaku community. Otaku are also known for their interest in niche or obscure games, which may not have a large mainstream audience but still have a dedicated fanbase (Ho et al., 2019).

In recent years, the influence of Otaku on the gaming industry has become more apparent. Companies have recognized the value of catering to this community, with many games featuring elements of anime and Japanese culture. Otaku have also become an important demographic for gaming influencers, as they are more likely to engage with content related to their interests (Ho et al., 2019).

Online influencers, including gaming influencers, have become a significant part of Otaku culture. Influencers are often seen as experts in their respective fields and are relied upon for their opinions and recommendations. Otaku may follow gaming influencers for a variety of reasons, such as staying up-to-date on the latest gaming news, learning about new games, or watching gameplay of their favorite games.

Prior studies have explored the role of online influencers in Otaku culture. A study by Hjorth and Chan (2009) found that gaming influencers have a significant impact on the purchase intentions of Otaku, as they are seen as credible sources of information. Another study by Törhönen et al. (2020) examined the relationship between gaming influencers and the brand loyalty of Otaku. The study found that Otaku who followed gaming influencers were more likely to be loyal to the brands recommended by the influencers. These studies suggest that gaming influencers have a significant influence on the behavior of Otaku, particularly in terms of their purchasing and brand loyalty.

Overall, these studies suggest that multiple dimensions of credibility contribute to the effectiveness of influencer marketing and that understanding and managing these dimensions is crucial for successful campaigns.

VTubers and streamers in the gaming industry

The gaming industry has witnessed significant growth in recent years, driven by advancements in technology and the increasing popularity of online platforms. As a result, the role of influencers in the gaming industry, particularly VTubers and streamers, has become more prominent. This subsection explores the impact of VTubers and streamers in the gaming industry and

highlights their unique characteristics and influence on the gaming community.

VTubers, or virtual YouTubers, are digital avatars controlled by content creators who use motion-capture technology to animate their characters. They often engage in various activities, including playing video games, live streaming, and interacting with their audience through social media platforms (Miyake, 2023). VTubers have gained popularity in the gaming industry as they offer a unique combination of animated characters, immersive storytelling, and engaging content, which appeals to a diverse audience, including the Otaku subculture (Berryman et al., 2021).

Streamers, on the other hand, are content creators who broadcast themselves playing video games on platforms such as Twitch, YouTube, and Facebook Gaming. These influencers engage with their audience through live chats, offering commentary, tips, and entertainment while playing the games (Hamilton et al., 2014). Streamers have become crucial in the gaming industry, as they help promote new games, create communities around specific titles, and drive engagement between game developers and players.

Both VTubers and streamers have significantly impacted the gaming industry by shaping consumer behavior, promoting games and gaming platforms, and influencing the development of new titles. As these influencers develop strong relationships with their audience (Hamilton et al., 2014), they can effectively influence purchasing decisions and create loyal fan bases for games and platforms (King & de la Hera, 2020; Sjöblom & Hamari, 2017). Moreover, their unique content and approach to gaming have diversified the gaming landscape, attracting new audiences and encouraging collaboration among content creators, game developers, and marketers.

In summary, VTubers and streamers play a critical role in the gaming industry, influencing consumer behavior, promoting games and platforms, and shaping the development of new titles. As their popularity continues to grow, understanding their impact and characteristics becomes essential for marketers, game developers, and content creators looking to capitalize on the opportunities offered by the gaming industry.

Method

Data are available on the Open Science Framework (https://osf.io/nms4q/?view_only=5ec0948a996042cc818a5627e708135d). A total of 686 female gamers, who follow at least one female gaming influencer on social media platforms, participated in this study. Participants were recruited via online gaming communities, social media platforms, and email invitations. Among them, 105 identified as Otaku, while 581 identified as non-Otaku. The mean age of the sample was 24.79 years old ($SD = 5.644$). Demographic information, including nationality, educational background, and occupation, was collected to provide a comprehensive understanding of the sample.

A survey was developed to measure variables of interest, including demographics, gaming behavior, and perceptions of two female gaming influencers: Aisha, representing VTubers, and Zbing Z, representing streamers. These influencers were selected for their popularity, diversity of content, and influence in their respective niches. The following variables were measured in this study adapted from previous studies (Boonchutima & Sankosik, 2022; Ohanian, 1990):

1. Expertise: Perceived expertise was assessed using a 5-point Likert scale ranging from 1 (not at all knowledgeable) to 5 (extremely knowledgeable). Five statements were used, with higher values indicating higher agreement and a higher level of perceived expertise. The scale exhibited acceptable internal consistency ($\omega = 0.88$ for Aisha and $\omega = 0.90$ for Zbing Z).
2. Attractiveness: Perceived attractiveness was evaluated using a 5-point Likert scale ranging from 1 (not at all attractive) to 5 (extremely attractive). Five statements were used, with higher values indicating higher agreement and a higher level of perceived attractiveness. The scale exhibited acceptable internal consistency ($\omega = 0.89$ for Aisha and $\omega = 0.88$ for Zbing Z).
3. Trustworthiness: Perceived trustworthiness was measured using a 5-point Likert scale ranging from 1 (not at all trustworthy) to 5 (extremely trustworthy). Five statements were used, with higher values indicating higher agreement and a higher level of perceived trustworthiness. The scale exhibited acceptable internal consistency ($\omega = 0.90$ for Aisha and $\omega = 0.90$ for Zbing Z).
4. Purchase intention: Purchase intention for a product recommended by the influencers was assessed using a 5-point Likert scale ranging from 1 (strongly disagree) to 5 (strongly agree). Three items were used, with higher values indicating higher agreement and a stronger purchase intention. The scale exhibited acceptable internal consistency ($\omega = 0.93$ for Aisha and $\omega = 0.96$ for Zbing Z).
5. Otaku classification: To be considered Otaku, participants had to meet three criteria: (1) play video

games for 5 hours or more daily, (2) collect items associated with virtual idols, and (3) attend events centered around their favorite anime, video game, or virtual idols. Participants not fulfilling these criteria were classified as non-Otaku.

This study was approved by the Faculty of Communication Arts and Master of Art in Communication's program committee at Chulalongkorn University. Invitations to participate in the survey were distributed via gaming forums. After reading the informed consent, participants who agreed to participate were asked a series of questions about influencers. Participants unfamiliar with the influencers were not asked to complete the main questionnaire. No financial incentives were provided as compensation for participation. Participants younger than 18 years were excluded from the final data analysis.

Descriptive statistics, correlation analyses, and regression analyses were conducted to examine relationships among the variables of interest. In presenting and interpreting our data, we aimed to follow the principles outlined by Valentine et al. (2015), which emphasize the interpretation of effect sizes. Regression analyses were conducted separately for the Otaku and non-Otaku groups to examine the predictive power of expertise, attractiveness, and trustworthiness on purchase intention. Effect sizes were reported along with the significance of the results. All statistical analyses were conducted using SPSS version 29.0.

Results

Respondent profile

The study included 686 respondents, with a mean age of 24.79 years (ranging from 18 to 80 years). Of the total respondents, 15.3% (105) were classified as Otaku and 84.7% (581) as non-Otaku. In terms of gender, the sample consisted of 381 (55.5%) males and 305 (44.5%) females. The majority of the respondents, both Otaku and non-Otaku, reported that they had seen game streaming (84% and 16%, respectively).

Table 1 presents a detailed view of the age distribution. The data analysis revealed a relatively young sample, with a mean age of 24.79 years and a standard deviation of 5.644. The age distribution is slightly positively skewed, with a skewness value of 2.173, indicating a higher concentration of respondents in the younger age group. The kurtosis value of 13.950 suggests a leptokurtic distribution, which is more peaked than a normal distribution. The percentiles demonstrate that 20% of respondents were 20 years or younger, 40% were 22.41 years or younger, 60% were 25.30 years or younger, and 80% were 29.17 years or younger.

Table 1. Respondents' ages.

	Otaku	Non-Otaku	Total
N	273	413	686
Mean	24.68	24.87	24.79
Standard error of mean	0.286	0.304	0.215
Median	24.17	23.57	23.87
Mode	25	20	20
Standard deviation	4.731	6.178	5.644
Variance	22.381	38.172	31.855
Skewness	0.812	2.504	2.173
Standard error of skewness	0.147	0.120	0.093
Kurtosis	0.810	15.758	13.950
Standard error of kurtosis	0.294	0.240	0.186
Range	27	62	62
Minimum	18	18	18
Maximum	45	80	80
Sum	6,737	10,271	17,008
Percentiles			
20	20.26	19.77	20.26
40	22.94	22.06	22.94
60	25.30	25.30	25.30
80	28.70	29.41	28.70

Furthermore, Otaku respondents had a slightly lower mean age (24.68 years) compared to non-Otaku respondents (24.87 years). The median age of Otaku respondents was 24.17 years, while the median age of non-Otaku respondents was 23.57 years. The mode age for Otaku respondents was 25 years, whereas the mode age for non-Otaku respondents was 20 years. The age range for all participants was between 18 and 80 years, with a range of 27 years for Otaku respondents and 62 years for non-Otaku respondents.

The standard deviation for age was 4.731 for Otaku respondents and 6.178 for non-Otaku respondents, indicating a larger variability in ages for non-Otaku participants. The variance for age was 22.381 for Otaku respondents and 38.172 for non-Otaku respondents. The distribution of ages for both groups was positively skewed, with Otaku respondents having a skewness of 0.812 and non-Otaku respondents having a skewness of 2.504. The kurtosis for Otaku respondents was 0.810, while the kurtosis for non-Otaku respondents was much higher at 15.758, indicating a greater degree of peakedness in the distribution of ages.

In conclusion, the age distribution analysis provides a comprehensive overview of the respondent profile for both Otaku and non-Otaku groups. The sample predominantly consisted of young individuals, mostly in their 20s. This information is crucial in understanding the preferences and behaviors of the target audience.

Table 2 and Figure 1 display the gender distribution of Otaku and non-Otaku respondents. Of the 686 participants, 55.5% identified as male and 44.5% identified as female. In the Otaku group, 29.9% identified as male and 52.1% as female, whereas in the non-Otaku group, 70.1% identified as male and 47.9% as female. This table highlights the differences in sex distribution between the two groups.

Table 3 and Figure 2 display the habits of respondents in relation to watching videos from VTubers and streamers and the duration of time spent watching videos.

For the frequency of watching videos from VTubers, 47.7% of Otaku respondents and 52.3% of non-Otaku respondents reported watching more than five times a week. Among Otaku respondents, the most common frequency reported was three or four times a week (73.3%), while for non-Otaku respondents, it was fewer than once a week (86.6%). For watching videos from streamers, 71.4% of Otaku respondents and 28.6% of non-Otaku respondents reported watching more than five times a week. Among Otaku respondents, the most common frequency reported was 3 or 4 times a week (78.4%), while for non-Otaku respondents, it was fewer than once a week (85.6%).

Regarding the duration of time spent watching videos, Otaku respondents reported longer durations than non-Otaku respondents. For example, 53.5% of Otaku respondents reported watching videos for 2 hours but less than 3 hours, compared to 46.5% of non-Otaku respondents. Among Otaku respondents, the most common duration reported was 2 hours but less than 3 hours, while for non-Otaku respondents, it was less than 1 hour.

Table 3 provides a summary of the habits of respondents in relation to watching videos from VTubers and streamers and the duration of time spent watching videos. The table highlights differences in frequency and duration of video-watching habits between Otaku and non-Otaku respondents.

Table 2. Respondents' gender.

		Otaku		Non-Otaku		Total	
Question	Options	*n*	%	*n*	%	*n*	%
Sex	Male	114	29.9	267	70.1	381	100.0
	Female	159	52.1	146	47.9	305	100.0

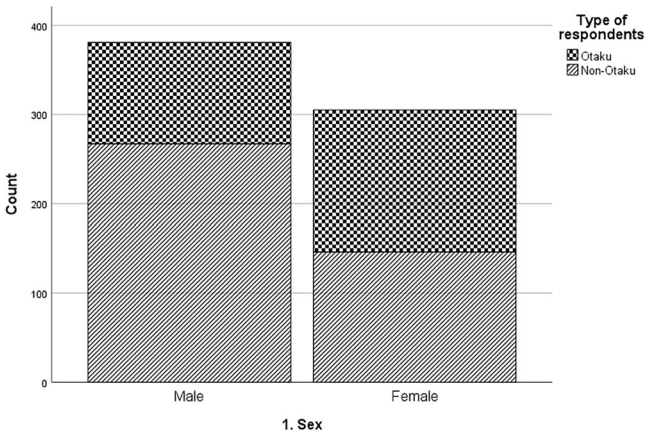

Figure 1. Gender and types of respondents.

Means comparison

Table 4 and Figure 3 display the means comparison between Otaku and non-Otaku respondents for variables related to expertise, attractiveness, trustworthiness, and purchase intention for both types of influencers: VTubers and streamers. The table also

Table 3. Habits of respondents.

		Otaku		Non-Otaku		Total	
Question	Options	*n*	%	*n*	%	*n*	%
Frequency in watching videos from VTubers	More than 5 times a week	41	47.7	45	52.3	86	100.0
	4–5 times a week	23	56.1	18	43.9	41	100.0
	3–4 times a week	74	73.3	27	26.7	101	100.0
	2–3 times a week	50	49.0	52	51.0	102	100.0
	1–2 times a week	51	50.0	51	50.0	102	100.0
	Fewer than 1 time a week	34	13.4	220	86.6	254	100.0
Frequency in watching videos from streamers	More than 5 times a week	50	71.4	20	28.6	70	100.0
	4–5 times a week	32	65.3	17	34.7	49	100.0
	3–4 times a week	80	78.4	22	21.6	102	100.0
	2–3 times a week	39	58.2	28	41.8	67	100.0
	1–2 times a week	27	31.8	58	68.2	85	100.0
	Fewer than 1 time a week	45	14.4	268	85.6	313	100.0
How long per session do you watch the videos?	Less than 1 hour	26	17.1	126	82.9	152	100.0
	1 hour but less than 2 hours	53	29.3	128	70.7	181	100.0
	2 hours but less than 3 hours	76	53.5	66	46.5	142	100.0
	3 hours but less than 4 hours	50	59.5	34	40.5	84	100.0
	4 hours but less than 5 hours	27	55.1	22	44.9	49	100.0
	5 hours but less than 6 hours	16	55.2	13	44.8	29	100.0
	6 hours or more	25	51.0	24	49.0	49	100.0

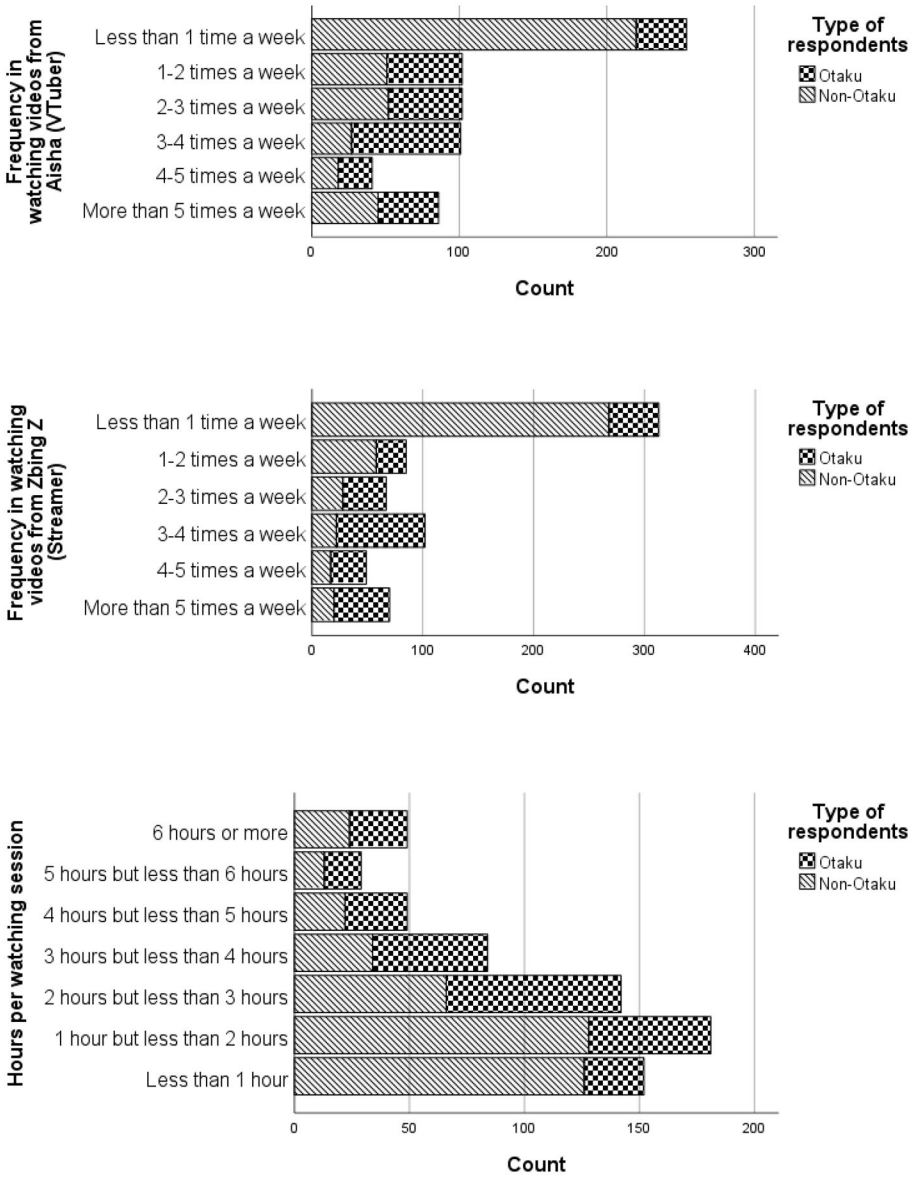

Figure 2. Habits.

provides the Cohen's d effect size for each comparison, which helps to interpret the magnitude and practical significance of the observed differences (Valentine et al., 2015).

For both types of influencers, Otaku respondents rated expertise (Cohen's $d = 0.502$) and trustworthiness (Cohen's $d = 0.438$) significantly higher than non-Otaku respondents, with medium effect sizes. Additionally, Otaku respondents had a higher purchase intention with a medium effect size (Cohen's $d = 0.532$). For VTubers, Otaku respondents rated expertise (Cohen's $d = 0.411$) and trustworthiness (Cohen's $d = 0.359$) significantly higher than non-Otaku respondents, with small to medium effect sizes. Additionally, Otaku respondents had a higher purchase intention with a small to medium effect size (Cohen's $d = 0.378$). For streamers, Otaku respondents rated expertise significantly higher than non-Otaku respondents, with a medium effect size (Cohen's $d = 0.518$). Additionally, Otaku respondents had a higher purchase intention, showing a medium to large effect size (Cohen's $d = 0.615$).

Table 4, therefore, highlights significant differences in perceptions of expertise, trustworthiness, and purchase intention between Otaku and non-Otaku respondents, with effect sizes ranging from small to medium and medium to large. In order to further examine the relationship between the factors influencing both types of respondents (Otaku and non-Otaku) and their perceptions of VTubers and streamers, a

Table 4. Means comparison between Otaku ($n = 105$) and non-Otaku ($n = 581$), and Cohen's d effect size.

Type of Influencer	Variables	Type of Respondents	*M*	*SD*	Cohen's *d*
Both types	Expertise	Otaku	3.54	0.640	0.502
		Non-Otaku	3.42	0.968	
	Attractiveness	Otaku	3.44	0.666	0.283
		Non-Otaku	3.43	0.991	
	Trustworthiness	Otaku	3.55	0.657	0.438
		Non-Otaku	3.50	0.987	
	Purchase intention	Otaku	2.93	0.836	0.532
		Non-Otaku	2.86	1.073	
VTuber	Expertise	Otaku	3.22	0.770	0.411
		Non-Otaku	3.13	1.034	
	Attractiveness	Otaku	3.47	0.789	0.287
		Non-Otaku	3.41	1.148	
	Trustworthiness	Otaku	3.43	0.798	0.359
		Non-Otaku	3.38	1.109	
	Purchase intention	Otaku	2.76	1.052	0.378
		Non-Otaku	2.77	1.154	
Streamer	Expertise	Otaku	3.87	0.766	0.518
		Non-Otaku	3.71	1.107	
	Attractiveness	Otaku	3.40	0.753	0.251
		Non-Otaku	3.45	1.050	
	Trustworthiness	Otaku	3.66	0.729	0.463
		Non-Otaku	3.61	1.072	
	Purchase intention	Otaku	3.10	0.949	0.615
		Non-Otaku	2.95	1.186	

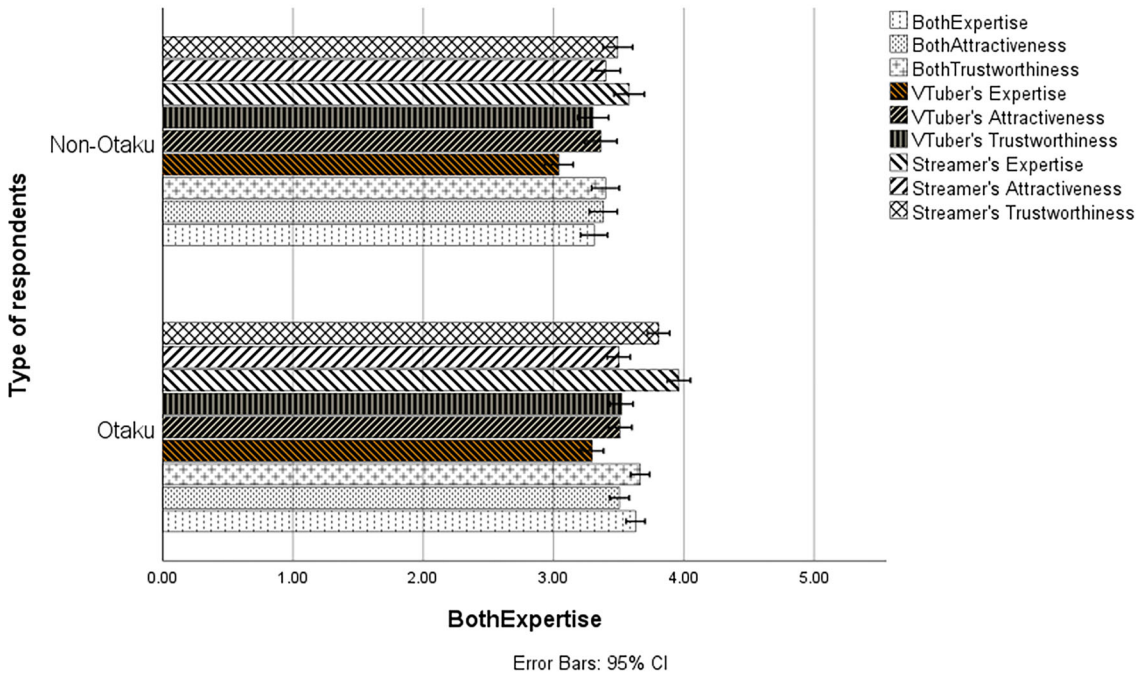

Figure 3. Means comparison.

regression analysis was conducted, the results of which are presented in Table 5.

Regression analysis

Table 5 and Figure 4 display the results of the regression analysis examining the impact of both types of influencers (VTubers and streamers) on the purchase intention of Otaku and non-Otaku respondents. For the combined model, which includes both types of influencers, expertise ($\beta = 0.330$), attractiveness ($\beta = 0.188$), and trustworthiness ($\beta = 0.237$) significantly predict the purchase intention of both respondent groups. The model accounts for 53.5% of the variance in purchase intention related to both types of influencers ($R^2 = 0.535$, adjusted $R^2 = 0.533$).

Table 5. Regression analysis of both types of influencers (VTubers and streamers) factors influencing the purchase intention of both types of respondents (Otaku and non-Otaku).

		Unstandardized coefficients		Standardized coefficients
Types of influencers	Model	B	SE	Beta
Both influencers[a]	(Constant)	−0.024	0.107	
	BothExpertise	0.370	0.076	0.330
	BothAttractiveness	0.206	0.079	0.188
	BothTrustworthiness	0.261	0.085	0.237
VTuber[b]	(Constant)	−0.066	0.098	
	VTuber's expertise	0.482	0.058	0.423
	VTuber's attractiveness	0.112	0.063	0.108
	VTuber's trustworthiness	0.276	0.069	0.258
Streamer[c]	(Constant)	0.133	0.122	
	Streamer's expertise	0.042	0.068	0.039
	Streamer's attractiveness	0.263	0.068	0.230
	Streamer's trustworthiness	0.492	0.078	0.438

Note: [a]$R = 0.732$, $R^2 = 0.535$, adjusted $R^2 = 0.533$, standard error of the estimate = 0.71035.
[b]$R = 0.757$, $R^2 = 0.572$, adjusted $R^2 = 0.571$, standard error of the estimate = 0.74595.
[c]$R = 0.683$, $R^2 = 0.467$, adjusted $R^2 = 0.465$, standard error of the estimate = 0.84372.

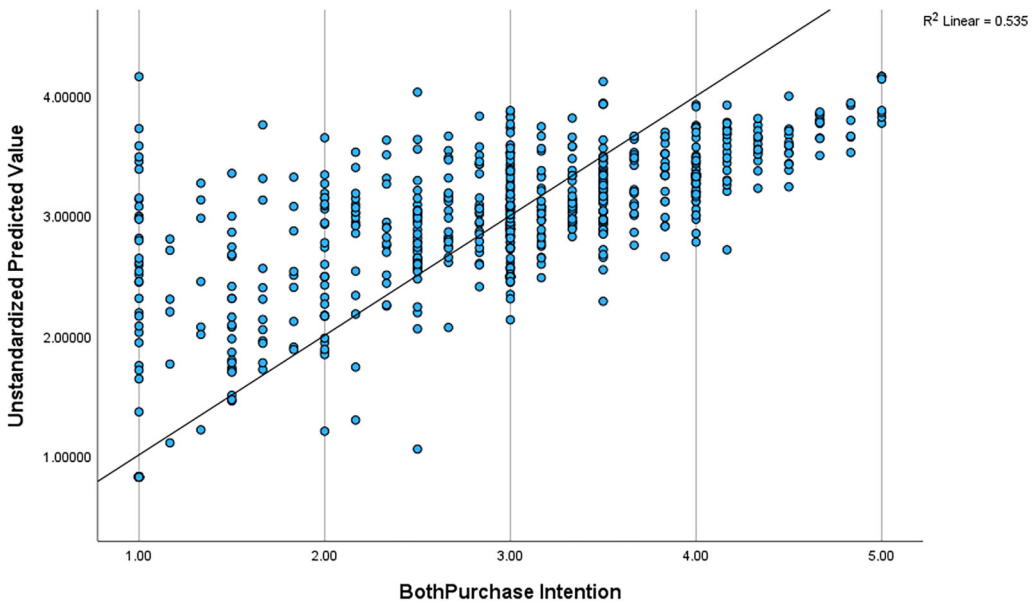

Figure 4. Regression model of both types of influencers that predict both types of respondents' purchase intention.

In the model focusing solely on VTubers (see also Figure 5), expertise ($\beta = 0.423$) and trustworthiness ($\beta = 0.258$) are significant predictors of purchase intention for both Otaku and non-Otaku respondents, while attractiveness ($\beta = 0.108$) is not. This model explains 57.2% of the variance in purchase intention related to VTubers ($R^2 = 0.572$, adjusted $R^2 = 0.571$).

For the model that includes only streamers (see also Figure 6), attractiveness ($\beta = 0.230$) and trustworthiness ($\beta = 0.438$) significantly predict the purchase intention of both respondent groups, whereas expertise ($\beta = 0.039$) is not a significant predictor. This model accounts for 46.7% of the variance in purchase intention related to streamers ($R^2 = 0.467$, adjusted $R^2 = 0.465$).

Overall, Table 6 demonstrates that expertise, attractiveness, and trustworthiness are significant predictors of purchase intention for both types of influencers (VTubers and streamers) among Otaku (Figure 7) and non-Otaku respondents (Figure 8). However, the influence of these predictors may vary depending on the influencer type. For Otaku, expertise and trustworthiness are the key factors driving purchase intention, while attractiveness plays a lesser role. In contrast, for non-Otaku, expertise and attractiveness emerge as the primary factors shaping purchase intention, with trustworthiness being an insignificant predictor.

In detail, Table 6 presents the results of the regression analysis evaluating the factors that influence the purchase intention of both Otaku and non-Otaku respondents in response to the combined effect of both types of influencers. For Otaku respondents, expertise ($\beta = 0.333$), trustworthiness ($\beta = 0.256$), and attractiveness ($\beta = 0.178$)

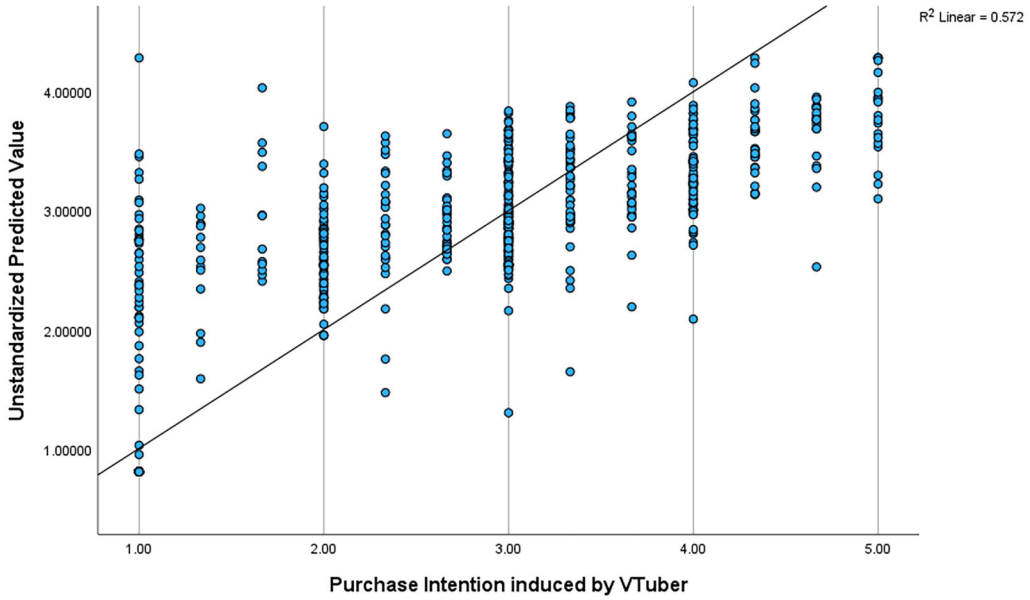

Figure 5. Regression model of the VTuber influencer that predicts both types of respondents' purchase intention.

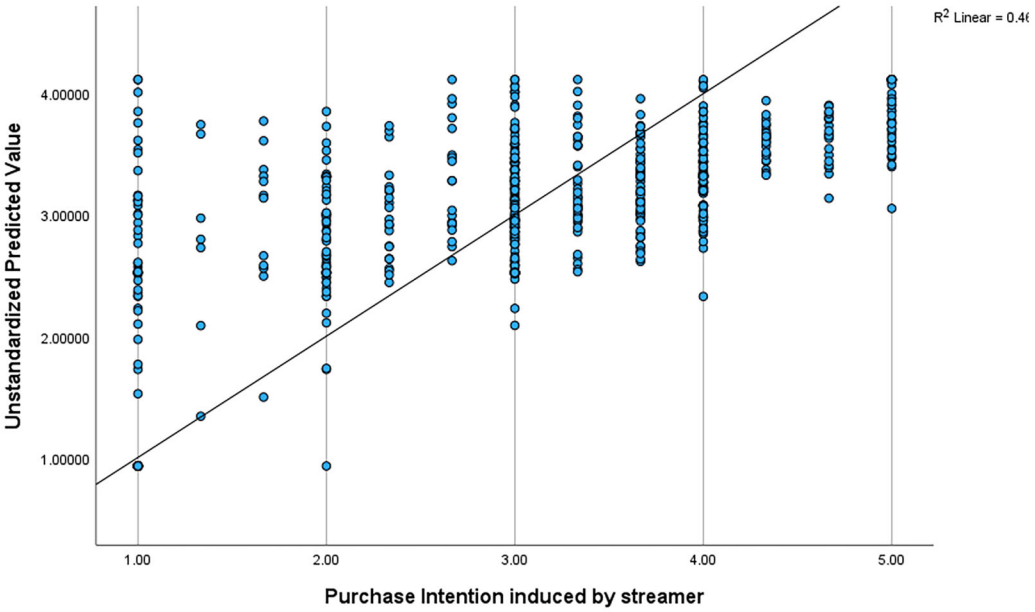

Figure 6. Regression model of the streamer influencer that predicts both types of respondents' purchase intention.

Table 6. Regression analysis of combined effect of two influencers' factors influencing the purchase intention of both types of respondents (Otaku vs. non-Otaku).

		Unstandardized coefficients		Standardized coefficients
Types of respondents	Model	B	SE	Beta
Otaku[a]	(Constant)	−0.581	0.225	
	BothExpertise	0.447	0.099	0.333
	BothAttractiveness	0.231	0.105	0.178
	BothTrustworthiness	0.342	0.109	0.256
Non-Otaku[b]	(Constant)	0.077	0.126	
	BothExpertise	0.278	0.109	0.261
	BothAttractiveness	0.306	0.113	0.297
	BothTrustworthiness	0.202	0.122	0.195

Note: [a]$R = 0.712$, $R^2 = 0.507$, adjusted $R^2 = 0.502$, standard error of the estimate = 0.57615.
[b]$R = 0.736$, $R^2 = 0.542$, adjusted $R^2 = 0.538$, standard error of the estimate = 0.77418.

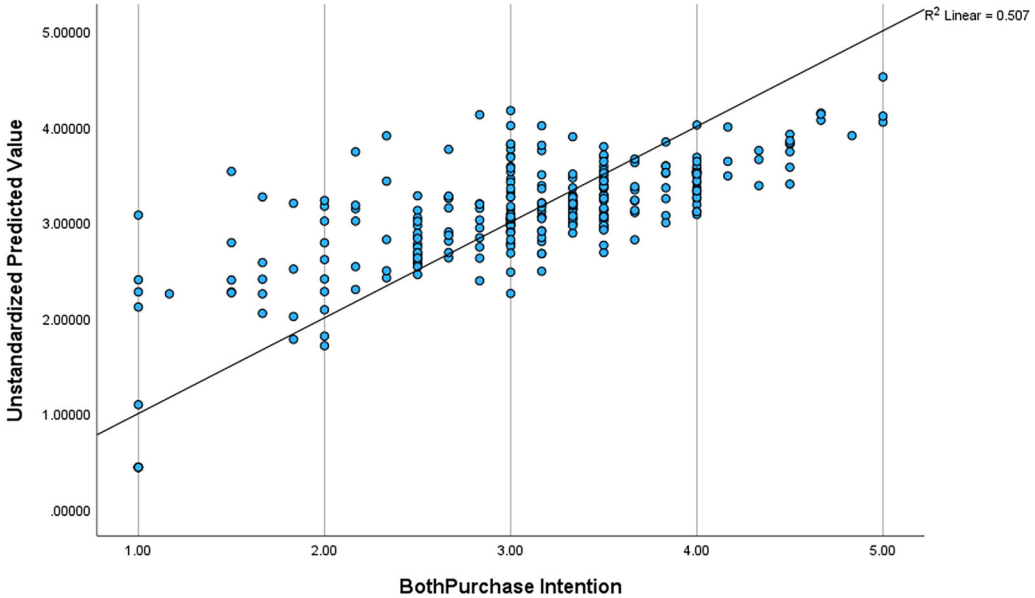

Figure 7. Regression analysis of both types of influencers influencing purchase intention of Otaku respondents.

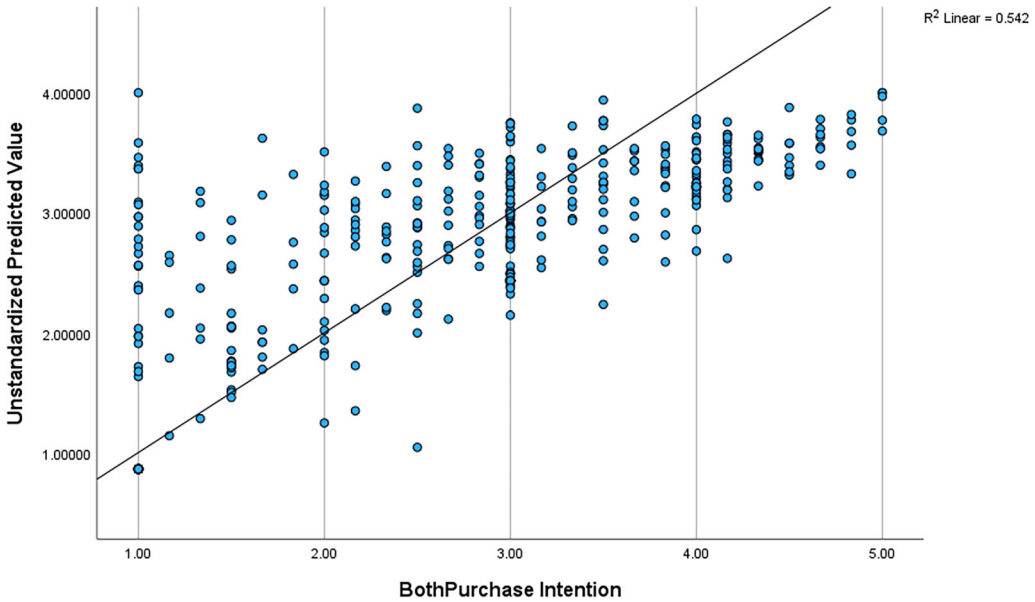

Figure 8. Regression analysis of both types of influencers influencing the purchase intention of non-Otaku respondents.

significantly predict their purchase intention related to the combined influence of both types of influencers. The model explains 50.7% of the variance in Otaku respondents' purchase intention ($R^2 = 0.507$, adjusted $R^2 = 0.502$). For non-Otaku respondents, attractiveness ($\beta = 0.297$), expertise ($\beta = 0.261$), and trustworthiness ($\beta = 0.195$) significantly impact their purchase intention related to the combined influence of both types of influencers, although trustworthiness is not a significant predictor. The model accounts for 54.2% of the variance in non-Otaku respondents' purchase intention ($R^2 = 0.542$, adjusted $R^2 = 0.538$).

Table 6 demonstrates that expertise, attractiveness, and trustworthiness are important factors in predicting the purchase intention of both Otaku and non-Otaku respondents when considering the combined effect of both types of influencers. However, the relative importance of these factors varies depending on the type of respondent. When examining Otaku respondents specifically, expertise has the most substantial impact on purchase intention, followed by trustworthiness and attractiveness.

This suggests that Otaku respondents are more likely to consider purchasing products or services promoted by influencers they perceive as knowledgeable

and skilled as well as trustworthy and attractive. For non-Otaku respondents, attractiveness has the strongest influence on purchase intention, followed by expertise and trustworthiness. This indicates that non-Otaku respondents place a higher value on the physical appeal of influencers when considering purchasing products or services they promote, in addition to their expertise and trustworthiness (Figures 9–11).

Table 7 presents the results of the regression analysis investigating the impact of VTuber influencers on the purchase intention of both Otaku and non-Otaku respondents. For both groups combined, expertise ($\beta = 0.423$) and trustworthiness ($\beta = 0.258$) significantly influence purchase intention, while attractiveness ($\beta = 0.108$) does not have a significant effect (see also Figure 8). The model explains 57.2% of the variance in respondents' perceptions of VTubers.

Focusing on Otaku respondents (see also Figure 9), expertise ($\beta = 0.398$) significantly influences their purchase intention related to VTubers, while attractiveness ($\beta = 0.186$) and trustworthiness ($\beta = 0.233$) do not have a significant impact. The model accounts for 57.3% of the variance in their purchase intention. For non-Otaku respondents (see also Figure 10), expertise ($\beta = 0.416$) and trustworthiness ($\beta = 0.255$) significantly affect their purchase intention related to VTubers, while attractiveness ($\beta = 0.118$) does not have a significant influence. The model explains 57.9% of the variance in non-Otaku respondents' purchase intention.

In summary, Table 7 demonstrates that expertise and trustworthiness are important predictors of purchase intention for both Otaku and non-Otaku respondents in relation to VTubers, while the impact of attractiveness varies between the two groups. In the case of Otaku respondents, attractiveness does not significantly influence their purchase intention related to VTubers ($\beta = 0.186$). This suggests that the perceived physical appeal of VTubers may not be as important for Otaku respondents when deciding whether to purchase products or services promoted by them.

Table 8 displays the results of the regression analysis examining the impact of streamers on the purchase intention of both Otaku and non-Otaku respondents. For both Otaku and non-Otaku respondents combined, trustworthiness ($\beta = 0.438$) and attractiveness ($\beta = 0.230$) significantly influence their purchase intention related to streamers. In simpler terms, respondents who find streamers more trustworthy and attractive are more likely to consider purchasing products or services they promote. The model explains 46.7% of the differences in respondents' purchase intention (see Figure 12).

When looking at Otaku respondents specifically, all three factors—trustworthiness ($\beta = 0.326$), expertise ($\beta = 0.211$), and attractiveness ($\beta = 0.207$)—significantly influence their purchase intention related to streamers. The model accounts for 44.8% of the variance in their purchase intention (Figure 13). For non-Otaku respondents, trustworthiness ($\beta = 0.471$), attractiveness ($\beta = 0.435$), and expertise ($\beta = -0.208$) significantly impact their purchase intention related to streamers. The model explains 48.7% of the

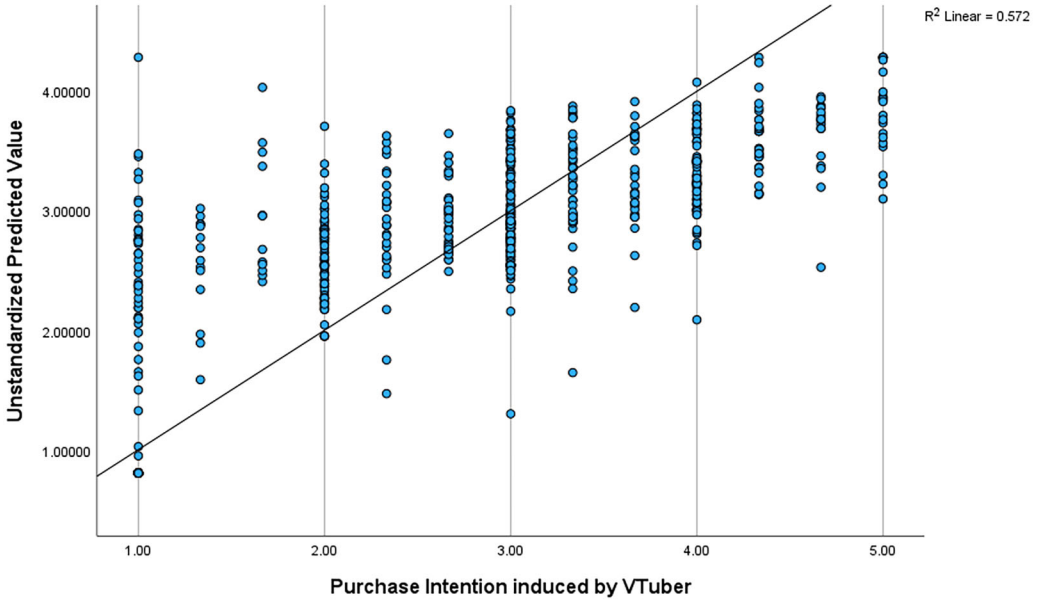

Figure 9. Regression analysis of the VTuber influencing the purchase intention of both types of respondents (Otaku and non-Otaku).

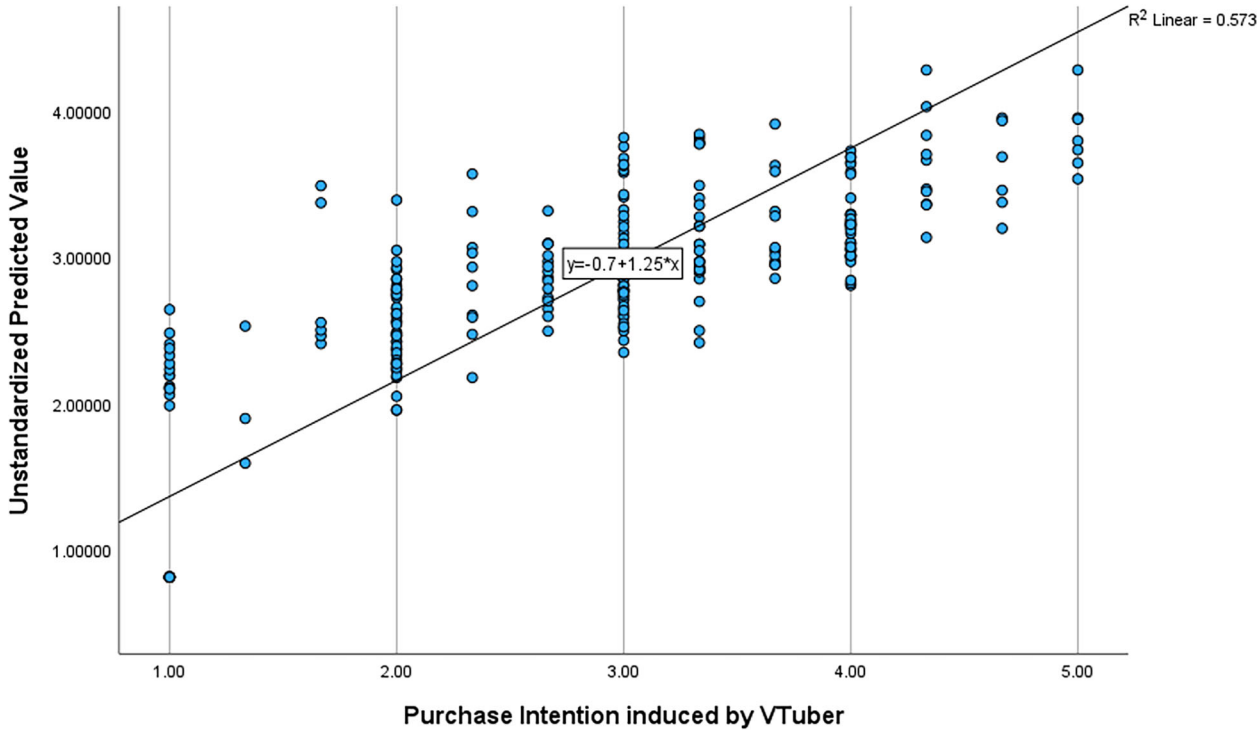

Figure 10. Regression analysis of the VTuber influencing the purchase intention of Otaku respondents.

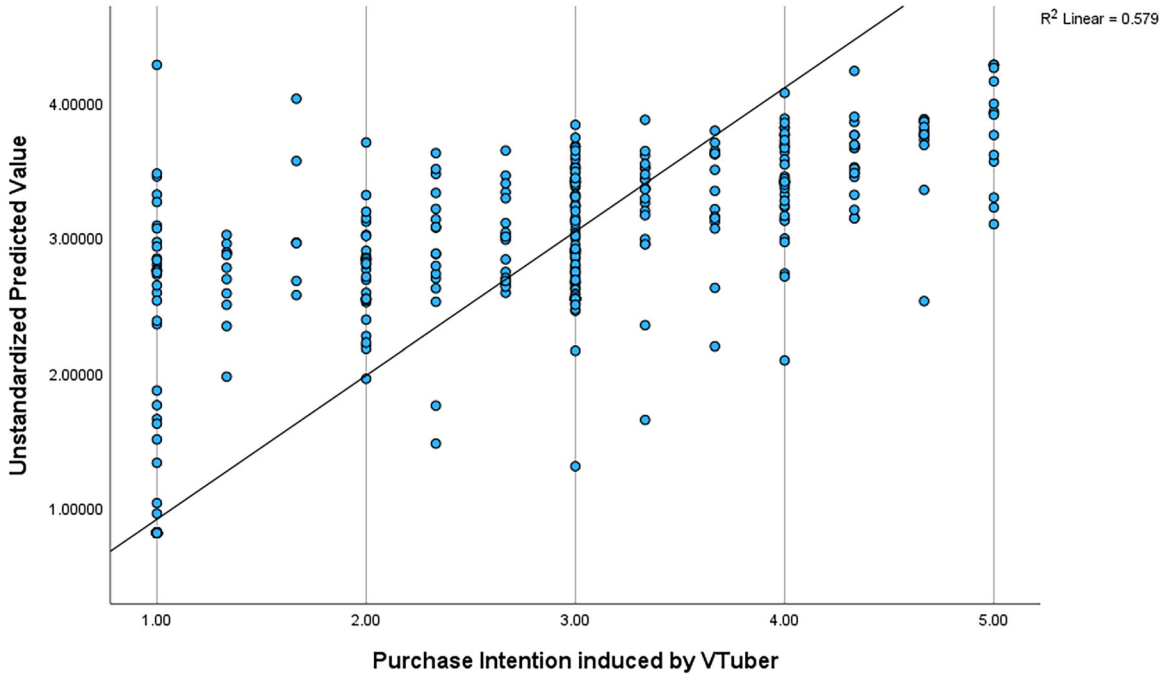

Figure 11. Regression analysis of the VTuber influencing the purchase intention of non-Otaku respondents.

differences in non-Otaku respondents' purchase intention (Figure 14).

Table 8 highlights that trustworthiness and attractiveness are key factors shaping the purchase intention of both Otaku and non-Otaku respondents when it comes to streamers. Expertise also plays a significant role, but its impact varies depending on the type of respondent. To provide further details, the influence of expertise on purchase intention related to streamers differs based on the respondent group. For Otaku

Table 7. Regression analysis of VTubers influencing the purchase intention of both types of respondents (Otaku and non-Otaku).

		Unstandardized coefficients		Standardized coefficients
Types of respondents	Model	*B*	Std. error	Beta
Both types of respondents[a]	(Constant)	−0.066	0.098	
	VTuber's expertise	0.482	0.058	0.423
	VTuber's attractiveness	0.112	0.063	0.108
	VTuber's trustworthiness	0.276	0.069	0.258
Otaku[b]	(Constant)	−0.899	0.318	
	VTuber's expertise	0.543	0.155	0.398
	VTuber's attractiveness	0.248	0.174	0.186
	VTuber's trustworthiness	0.307	0.168	0.233
Non-Otaku[c]	(Constant)	0.091	0.116	
	VTuber's expertise	0.449	0.076	0.416
	VTuber's attractiveness	0.113	0.081	0.118
	VTuber's trustworthiness	0.252	0.093	0.255

Note: [a]$R = 0.757$, $R^2 = 0.572$, adjusted $R^2 = 0.572$, standard error of the estimate = 0.74595.
[b]$R = 0.757$, $R^2 = 0.573$, adjusted $R^2 = 0.569$, standard error of the estimate = 0.64859.
[c]$R = 0.761$, $R^2 = 0.579$, adjusted $R^2 = 0.576$, standard error of the estimate = 0.79369.

Table 8. Regression analysis of streamer influencing the purchase intention of both types of respondents (Otaku and non-Otaku).

		Unstandardized coefficients		Standardized coefficients
Types of respondents	Model	B	Std. error	Beta
Both types of respondents[a]	(Constant)	0.133	0.122	
	Streamer's expertise	0.042	0.068	0.039
	Streamer's attractiveness	0.263	0.068	0.230
	Streamer's Trustworthiness	0.492	0.078	0.438
Otaku[b]	(Constant)	−0.404	0.255	
	Streamer's expertise	0.271	0.088	0.211
	Streamer's attractiveness	0.268	0.086	0.207
	Streamer's Trustworthiness	0.441	0.104	0.326
Non-Otaku[c]	(Constant)	0.238	0.139	
	Streamer's expertise	−0.211	0.097	−0.208
	Streamer's attractiveness	0.462	0.102	0.435
	Streamer's Trustworthiness	0.490	0.107	0.471

Note. [a]$R = 0.683$, $R^2 = 0.467$, adjusted $R^2 = 0.465$, standard error of the estimate = 0.84372.
[b]$R = 0.669$, $R^2 = 0.448$, adjusted $R^2 = 0.441$, standard error of the estimate = 0.71552.
[c]$R = 0.698$, $R^2 = 0.487$, adjusted $R^2 = 0.484$, standard error of the estimate = 0.87989.

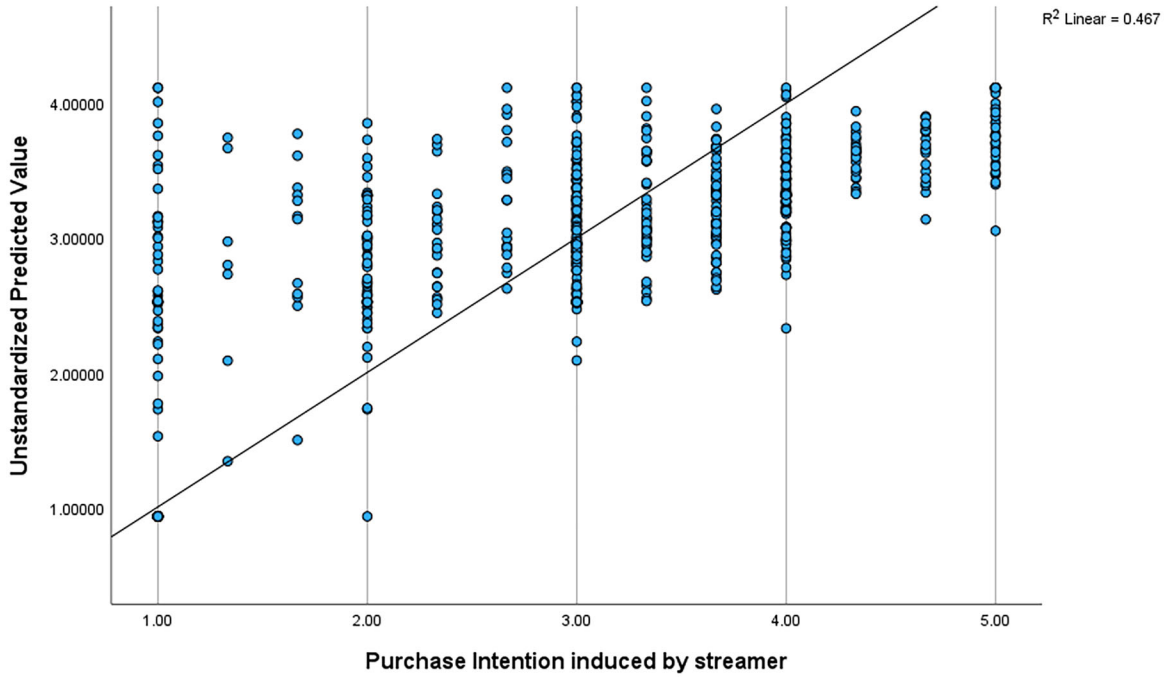

Figure 12. Regression analysis of the streamer influencing the purchase intention both types of respondents.

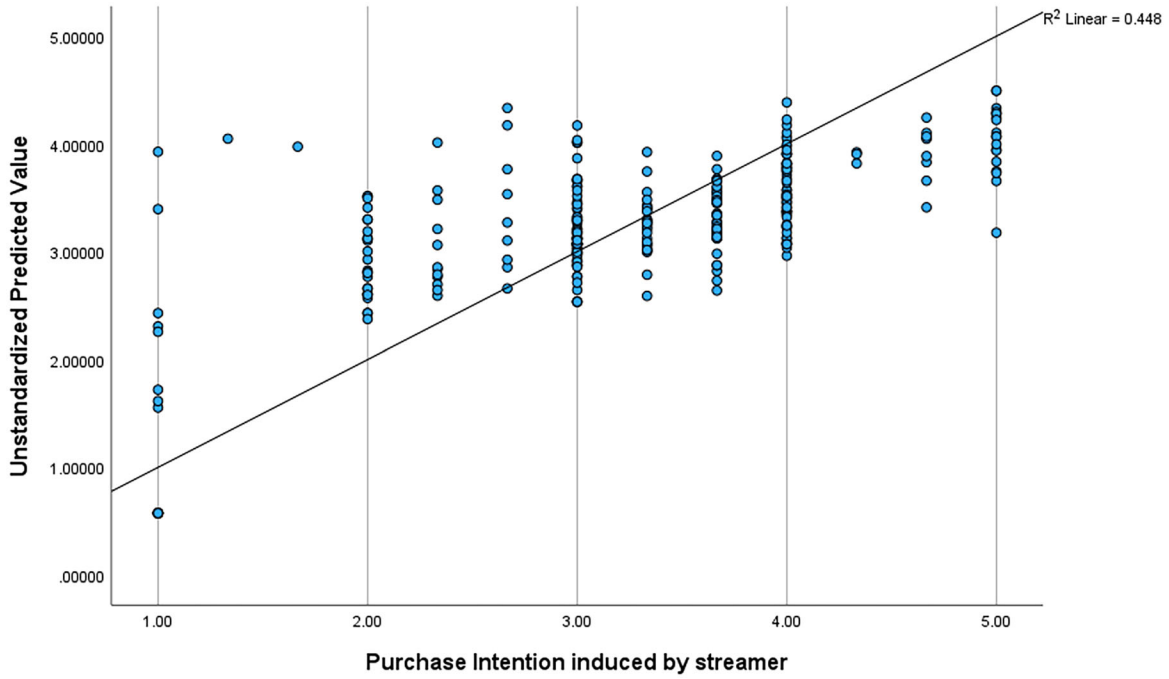

Figure 13. Regression analysis of the streamer influencing the purchase intention of Otaku respondents.

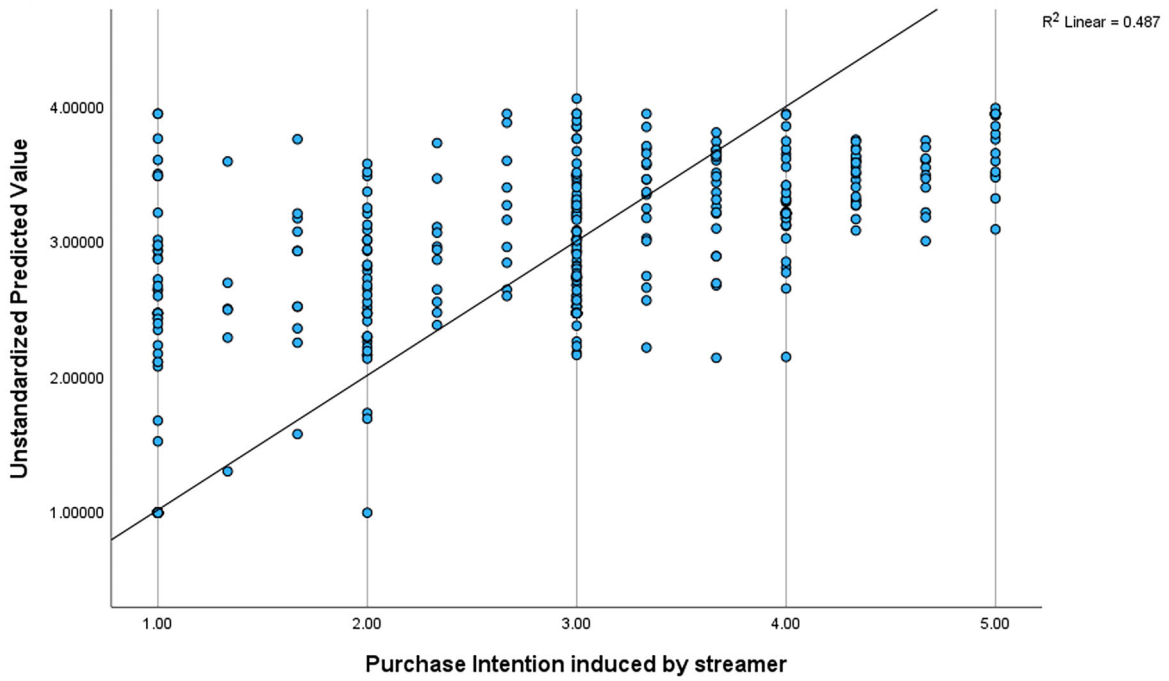

Figure 14. Regression analysis of the streamer influencing the purchase intention of non-Otaku respondents.

respondents, expertise has a positive effect on purchase intention ($\beta = 0.211$), meaning that when they perceive streamers as more knowledgeable and skilled, they are more likely to consider purchasing products or services promoted by them. In contrast, for non-Otaku respondents, expertise has a negative effect on purchase intention ($\beta = -0.208$). This suggests that when non-Otaku respondents perceive streamers as more knowledgeable and skilled, they are less likely to consider purchasing products or services promoted by them. This unexpected finding may imply that non-Otaku respondents prefer streamers who are more relatable or approachable, rather than those who exhibit a high level of expertise.

Summary of the findings

The study investigates the influence of VTubers and streamers on the purchase intentions of Otaku and non-Otaku respondents, with a specific focus on the roles of expertise, attractiveness, and trustworthiness. The findings reveal that both Otaku and non-Otaku respondents perceive VTubers and streamers differently in terms of expertise, attractiveness, and trustworthiness. Otaku respondents tend to favor VTubers, who they perceive as having higher expertise and trustworthiness than streamers, while non-Otaku respondents prefer streamers, who they perceive as having higher attractiveness.

Regardless of the type of influencer, both Otaku and non-Otaku respondents consider expertise, attractiveness, and trustworthiness as important factors when making purchase decisions based on influencer endorsements. However, the significance of these factors varies depending on the type of influencer and respondent. VTubers are generally perceived as having higher levels of expertise and trustworthiness, while streamers are seen as more attractive.

The study consistently shows that expertise, attractiveness, and trustworthiness are significant predictors of purchase intention for both types of influencers and both respondent types. These factors play a crucial role in determining the impact of VTubers and streamers on the purchase intentions of their respective audiences. Brands targeting Otaku and non-Otaku audiences should take these findings into account when selecting influencers to partner with and developing marketing strategies that cater to the preferences of their target demographics.

Discussion

The present study adds to the existing body of knowledge on influencer marketing, source credibility, and the Otaku subculture. The regression analysis results presented in Tables 5 through 8 align with previous research that emphasized the importance of expertise, attractiveness, and trustworthiness in shaping perceptions of influencers (Ohanian, 1990; Petty & Cacioppo, 1986).

Expertise is a significant predictor for both types of influencers. According to the elaboration likelihood model, individuals who perceive an influencer as an expert are more likely to be persuaded by the influencer's message (Petty & Cacioppo, 1986). In the context of this study, non-Otaku respondents may not fully understand the content produced by VTubers, which might impact their perception of expertise (Table 8). This finding highlights the importance of content familiarity when assessing expertise.

Attractiveness is a significant predictor for both types of influencers, as it can impact how viewers perceive them. According to the halo effect, individuals often attribute positive qualities to attractive people (Dion et al., 1972). In this study, streamers rely more on their physical appearance to attract viewers, while VTubers depend on their virtual avatar's appearance and interaction with viewers (Tables 7 and 8). This finding emphasizes the importance of influencers considering the visual appeal of their content and appearance. However, the impact of attractiveness may vary depending on the context. For example, Bhogal et al. (2016) found that attractiveness did not have a direct relationship with altruism in two modified dictator games, suggesting that the role of attractiveness is complex and context-dependent. Further research is needed to better understand the various ways attractiveness influences both perceptions and behaviors across different social contexts and situations.

Trustworthiness plays a crucial role in shaping purchase intention due to viewing both types of influencers. Source credibility theory posits that individuals are more likely to be persuaded by a message from a trustworthy source (Ohanian, 1990). Non-Otaku respondents seem to place greater emphasis on trustworthiness when forming purchase intention induced by streamers (Table 8), potentially due to their lack of prior relationships with these influencers. This finding highlights the importance of building trust with the audience, particularly for streamers targeting non-Otaku viewers.

This study expands on the existing literature by examining the impact of VTubers and streamers on Otaku and the influence of expertise, attractiveness, and trustworthiness on Otaku and non-Otaku respondents' purchase intention. The findings emphasize the importance of expertise, attractiveness, and trustworthiness in shaping purchase intention induced by influencers. Additionally, the study shows that Otaku respondents, especially when it comes to VTubers, tend to place more importance on expertise when forming their purchase intentions. As a result, understanding the content and preferences of the target audience, particularly for VTubers targeting Otaku respondents, becomes crucial to ensure that the influencer's expertise aligns with the audience's expectations. Influencers should consider these factors when creating content to maximize their impact on different audiences.

Limitations and future directions

This study has several limitations. The sample was limited to Thai respondents, which may affect the generalizability of the findings to other cultural contexts. Future research should include diverse cultural backgrounds for a comprehensive understanding of influencer impact.

The study relied on self-reported survey data, which could be subject to biases. A mixed-methods approach, incorporating interviews, observations, or content analysis, could help validate the survey data. The study focused on gaming VTubers and streamers, overlooking other influencer types such as bloggers, celebrities, and micro-influencers as well as other genres of content. Expanding the scope of future research could provide a more nuanced understanding of factors shaping consumer perceptions and behaviors.

The study did not examine specific content shared by influencers, even within the gaming category. Investigating the influence of different content types and genres on consumer outcomes is crucial for future research. Moreover, the study focused on influencers' role in shaping consumers' perceptions and behaviors, neglecting other potential factors such as social norms, personal values, and prior experiences. Exploring the interplay between influencers and these factors is essential for understanding the complex nature of consumers' decision-making processes.

Theoretical implications

The findings of this study contribute to the existing literature on influencer marketing, source credibility, and Otaku subculture. By examining the impact of expertise, attractiveness, and trustworthiness on the perception of VTubers and streamers among Otaku and non-Otaku respondents, the study expands our understanding of how different factors influence consumer perceptions. Furthermore, the study highlights the role of Otaku subculture in shaping consumers' perceptions of influencers, providing a new perspective on the interplay between cultural factors and influencer marketing.

Practical implications

For marketers and brand managers, the study offers valuable insights into developing effective influencer marketing strategies. The results suggest that expertise, attractiveness, and trustworthiness should be considered when selecting influencers to maximize campaign effectiveness. Moreover, understanding the differences in purchase intention between Otaku and non-Otaku respondents can help marketers tailor their strategies based on the characteristics of their target audience. By carefully selecting influencers that align with their brand and target audience, marketers can enhance the success of their influencer marketing campaigns.

Content creators and influencers can also benefit from the study's findings. The results emphasize the importance of expertise, attractiveness, and trustworthiness in shaping audience's purchase intention. VTubers should focus on demonstrating their expertise and trustworthiness, while streamers should prioritize showcasing their attractiveness and trustworthiness. By enhancing these attributes, influencers can better connect with their audience, ultimately leading to increased engagement, credibility, and influence in their respective niches.

Conclusion

This study explored the impact of source credibility on the perception of VTuber and streamer influencers by Otaku and non-Otaku respondents in Thailand. The findings of this study reveal that expertise is a significant predictor of the purchase intention induced by VTubers for both Otaku and non-Otaku respondents, while attractiveness is not a significant predictor for non-Otaku respondents. For streamers, both attractiveness and trustworthiness significantly predict the purchase intention of both Otaku and non-Otaku respondents, while expertise is not a significant predictor.

Furthermore, the results also demonstrate that the purchase intention induced by both types of influencers (VTubers and streamers) may vary depending on the type of respondents (Otaku vs. non-Otaku). Additionally, this study contributes to the body of knowledge on source credibility, social psychology, and influencer marketing. Future research can address the limitations of this study and explore other factors that may influence the perception of influencers by different types of audiences.

Overall, the results of this study provide important insights for marketers and businesses interested in leveraging influencer marketing in Thailand. By understanding the factors that influence the perception of influencers by different types of audiences, businesses can develop more effective influencer marketing strategies that resonate with their target audience.

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Disclosure statement

No potential conflict of interest was reported by the author(s).

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Data availability statement

Data, headlines, and additional online materials are openly available at the project's Open Science Framework page (https://osf.io/nms4q/?view_only=5ec0948a996042cc818a5627e708135d).

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